INCH-POUND
MS27952B
22 February 2022
SUPERSEDING
MS27952A
22 April 1968

# DETAIL SPECIFICATION SHEET

# NUT, PLAIN-HEXAGON, JAM, LEFT HAND

Reactivated after 22 February 2022 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the described herein shall consist of this specification sheet and commercial specifications ASME B18.2.2 and ASTM A563/A563M.

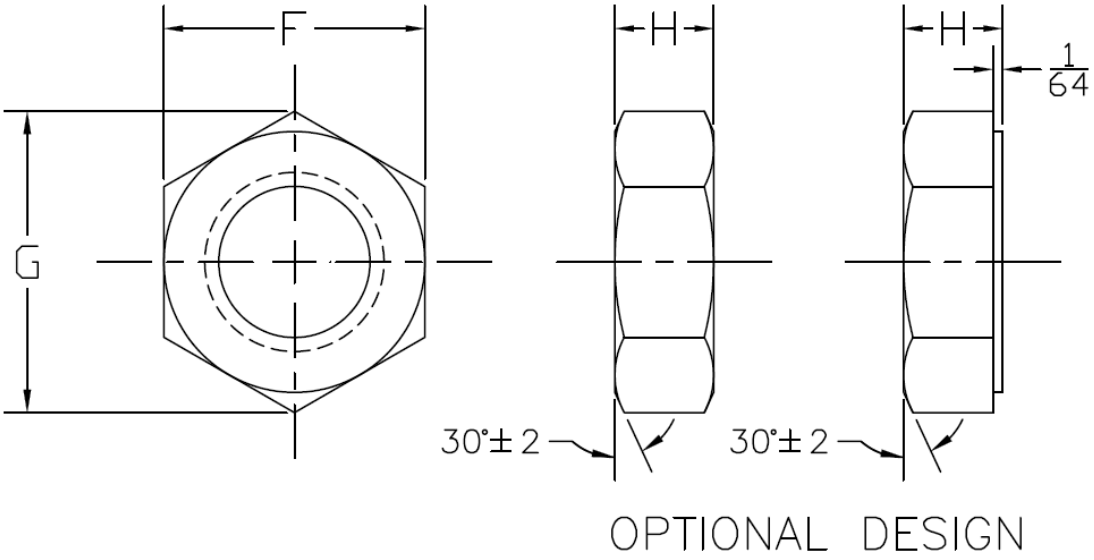



FIGURE 1. Nut, plain-hexagon, jam, left hand.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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REQUIREMENTS:

1. Material. Unless otherwise specified, carbon and alloy steel nuts shall be made per the chemical and mechanical requirements of ASTM A563/A563M Grade A.

2. Protective finish: Zinc-coated (hot dipped galvanize) in accordance with ASTM A153/A153M, Surface treatment of hot dip galvanized with any chromate free process in accordance with ASTM D6386.

3. Threads. The threads shall be left-hand in accordance with ASME B1.1.

4. Conformance. Certificates of compliance to paragraph 1, 2, and 3 shall be provided to verify conformance.

5. Part or Identifying Number (PIN). The PIN consists of the following:

MS27952 – 1

- 1 — Dash number from Table I indicating size/dimensions (cross-referenced to Figure 1.)
- MS27952 — Specification Sheet number

Example: MS27952-1 is a Nominal 1/4 inch plain-hexagon left hand jam nut, 5/32 inch thickness with galvanized zinc coating.

TABLE I. Dash numbers and dimensions (inch).

| DASH NUMBER | NOMINAL (NOM) SIZE OR BASIC MAJOR DIAMETER OF THREADS | | T PITCH & THREAD DESIGNATION LEFT HAND | F WIDTH ACROSS FLATS | | | C WIDTH ACROSS CORNERS | | H THICKNESS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | NOM | MAX | MIN | MAX | MIN | NOM | MAX | MIN |
| 1 | 1/4 | 0.2500 | 20 UNC-2B-LH | 7/16 | 0.4375 | 0.428 | 0.505 | 0.488 | 5/32 | 0.1625 | 0.150 |
| 30 | 1/4 | 0.2500 | 28 UNF-2B-LH | 7/16 | 0.4375 | 0.428 | 0.505 | 0.488 | 5/32 | 0.1630 | 0.150 |
| 3 | 5/16 | 0.3125 | 18 UNC-2B-LH | 1/2 | 0.5000 | 0.489 | 0.577 | 0.557 | 3/16 | 0.1950 | 0.180 |
| 31 | 5/16 | 0.3125 | 24 UNF-2B-LH | 1/2 | 0.5000 | 0.489 | 0.577 | 0.557 | 3/16 | 0.1950 | 0.180 |
| 5 | 3/8 | 0.3750 | 16 UNC-2B-LH | 9/16 | 0.5625 | 0.551 | 0.650 | 0.628 | 7/32 | 0.2770 | 0.210 |
| 32 | 3/8 | 0.3750 | 24 UNF-2B-LH | 9/16 | 0.5625 | 0.551 | 0.650 | 0.628 | 7/32 | 0.2770 | 0.210 |
| 33 | 7/16 | 0.4375 | 20 UNF-2B-LH | 11/16 | 0.6875 | 0.675 | 0.794 | 0.768 | 1/4 | 0.2600 | 0.240 |
| 7 | 1/2 | 0.5000 | 18 UNC-2B-LH | 3/4 | 0.7500 | 0.736 | 0.866 | 0.840 | 5/16 | 0.3230 | 0.302 |
| 34 | 1/2 | 0.5000 | 20 UNF-2B-LH | 3/4 | 0.7500 | 0.736 | 0.866 | 0.840 | 5/16 | 0.3230 | 0.302 |
| 9 | 5/8 | 0.6250 | 11 UNC-2B-LH | 15/16 | 0.9375 | 0.922 | 1.083 | 1.051 | 3/8 | 0.3870 | 0.363 |
| 35 | 5/8 | 0.6250 | 18 UNF-2B-LH | 15/16 | 0.9375 | 0.922 | 1.083 | 1.051 | 3/8 | 0.3870 | 0.363 |
| 11 | 3/4 | 0.7500 | 10 UNC-2B-LH | 1 1/8 | 1.1250 | 1.088 | 1.299 | 1.240 | 27/64 | 0.4460 | 0.398 |
| 13 | 7/8 | 0.8750 | 9 UNC-2B-LH | 1 1/8 | 1.1250 | 1.269 | 1.516 | 1.447 | 31/64 | 0.5100 | 0.458 |
| 15 | 1 | 1.0000 | 8 UNC-2B-LH | 1 1/2 | 1.5000 | 1.450 | 1.732 | 1.653 | 35/64 | 0.5750 | 0.519 |
| 36 | 1 | 1.0000 | 12 UNF-2B-LH | 1 1/2 | 1.5000 | 1.450 | 1.732 | 1.653 | 35/64 | 0.5750 | 0.519 |
| 17 | 1 1/8 | 1.1250 | 7 UNC-2B-LH | 1 11/16 | 1.6875 | 1.631 | 1.949 | 1.859 | 39/64 | 0.6390 | 0.579 |
| 19 | 1 1/4 | 1.2500 | 7 UNC-2B-LH | 1 23/40 | 1.5750 | 1.812 | 2.165 | 2.066 | 23/32 | 0.7510 | 0.687 |
| 37 | 1 1/4 | 1.2500 | 12 UNF-2B-LH | 1 23/40 | 1.5750 | 1.812 | 2.165 | 2.066 | 23/32 | 0.7510 | 0.687 |
| 21 | 1 1/2 | 1.5000 | 6 UNC-2B-LH | 2 1/4 | 2.2500 | 2.175 | 2.588 | 2.480 | 27/32 | 0.8800 | 1.808 |
| 23 | 1 3/4 | 1.7500 | 6 UNC-2B-LH | 2 5/8 | 2.6250 | 2.538 | 3.031 | 2.893 | 31/32 | 1.0090 | 0.929 |
| 25 | 2 | 2.000 | 4 ½ UNC-2B-LH | 3 | 3.0000 | 2.900 | 3.464 | 3.306 | 1 3/32 | 1.1380 | 1.050 |
| 27 | 2 1/2 | 2.5000 | 4 UNC-2B-LH | 3 3/4 | 3.7500 | 3.625 | 4.330 | 4.133 | 1 29/64 | 1.5050 | 1.401 |

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Table II. Interchangeability Table for Cancelled Parts

Dash numbers for the following Bearing surface type of jam nut, as given in the original issue of this standard, are cancelled after 22 April 1968. The cancelled nuts should be used until existing stock is depleted. Use only the superseding nuts for design and replacement.

| PART NUMBER | |
|---|---|
| CANCELLED | SUPERSEDED BY |
| DASH NUMBER | DASH NUMBER |
| 2<br>4<br>6<br>8<br>10 | 1<br>3<br>5<br>7<br>9 |
| 12<br>14<br>16<br>18<br>20 | 11<br>13<br>15<br>17<br>19 |
| 22<br>24<br>26<br>28 | 21<br>23<br>25<br>27 |

6 Dimensions. Unless otherwise specified, all dimensions are in inches.

7 Design. For design feature purposes, this document takes precedence over all other documents.

8 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids. Copies of ASTM specifications are available online at https://www.astm.org/Standard/standards-and-publications. Copies of MIL/FED documents are available online at https://quicksearch.dla.mil. Copies of SAE specifications are available online at https://www.sae.org/standards.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH
DLA – GS

Reviewer:
Army - AR

Preparing Activity:
Army – AT

(Project 5310-2022-008)

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